Hybrid CNN–Transformer Networks with Clinical Metadata for Skin Lesion Classification

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Declaration

It is hereby declared that :

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Abstract

Dermoscopy technique for classification of skin lesions may be useful in early diagnosis of severe skin diseases and in the clinical management of skin lesions. The challenge is to create an accurate and clinically usable system. We create two versions of this design, based on requirements. The first one is light in weight. It adopts a truncated MobileNetV2 backbone along with a compact transformer, and with only 3.98 million parameters, it achieves 60.8% BMA (balanced multi-class accuracy). This compact design facilitates easy installation within typical clinic environments, where space is a constraint. The second version emphasizes more balanced accuracy than capacity of the parameters. It is composed of two different pre-trained backbones (DenseNet-121 and EfficientNet-B0) and a learned attention mechanism that selectively fuses information from both followed with a Kolmogorov Arnold network as the final classifier and attains a BMA of 64.4% on 16.45M parameters. A dermatologist might consider the clinical context in the classification process but this information is seldom taken into account by image-based classifiers. In this case, both models include patient metadata as added diagnostic context directly into the classification pipeline with the data for age, sex and location of the body part where the lesion occurred. The models provide an efficient and accurate balance for real-world dermatology applications over same-budget baselines.

Keywords: Skin lesion classification; ISIC-2019; dermoscopy; CNN–Transformer; metadata fusion; cross-attention; Kolmogorov–Arnold Network; class imbalance; balanced multi-class accuracy; lightweight deep learning.

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Chapter 1

Introduction

1.1 Background

Skin cancer is among the most common cancers worldwide, and the prognosis for a patient depends heavily on how early the lesion is found. Melanoma in particular is highly treatable when caught early but becomes far more dangerous once it spreads. Dermoscopy a non-invasive imaging technique that removes surface glare and reveals the sub-surface structure of a lesion has become a standard tool for this kind of early assessment. Reading dermoscopic images well, however, takes years of training, and even experienced dermatologists do not always agree with one another. This gap between the volume of lesions that need screening and the number of specialists available to read them is what motivates automated, image-based diagnostic support.

Deep learning has reshaped this area over the past decade. Convolutional neural networks (CNNs) trained on large dermoscopic collections reached dermatologist-level performance on melanoma recognition [5], and the annual ISIC challenges have since turned skin lesion classification into a well-defined benchmark task. The ISIC-2019 dataset, which we use throughout this work, contains 25,331 labelled dermoscopic images spanning eight diagnostic categories: melanoma (MEL), melanocytic nevus (NV), basal cell carcinoma (BCC), actinic keratosis (AK), benign keratosis (BKL), dermatofibroma (DF), vascular lesion (VASC), and squamous cell carcinoma (SCC).

More recently, Vision Transformers (ViTs) [21] and hybrid designs that place a transformer on top of convolutional features [23], [24] have shown that modelling the global structure of a lesion its overall asymmetry, the regularity of its border complements the local texture that convolutions capture so well. These two views of a lesion, the local and the global, are exactly the cues a clinician is trained to weigh, which makes the CNN–Transformer combination a natural fit for dermoscopy.

1.2 Rationale of the Study or Motivation

Two practical problems keep most published skin-lesion models from being used in an everyday clinic. The first is cost. The strongest results on ISIC-2019 come from large ensembles of heavy networks [15], [19]; these are accurate but impractical to run on the modest hardware found in a typical dermatology practice or on a portable device next to the patient. The second is that image-only models ignore information a dermatologist always has in front of them. The age of the patient, their sex, and the part of the body where the lesion sits all shift the prior probability of each diagnosis a scaly patch on the face of an elderly patient is read very differently from the same patch on a young person's back. This clinical context is recorded in the ISIC-2019 metadata, yet it is routinely discarded by classifiers that look only at pixels.

Our study is driven by the belief that these two problems should be solved together rather than separately. We argue first that a carefully designed lightweight hybrid can match or beat much larger image-only baselines on balanced accuracy while staying small enough to deploy, and second that the clinical metadata a dermatologist already has on hand can be folded in at negligible cost as additional diagnostic context, handled so that missing fields are never imputed. We also recognise that not every deployment has the same constraints, so we deliver two models from one design philosophy: a genuinely compact model for constrained settings and a higher-capacity model for situations where balanced accuracy matters more than footprint.

1.3 Problem Statement

ISIC-2019 is severely imbalanced. The most common class, nevus, has roughly 12,875 images, while dermatofibroma and vascular lesions have only about 239 and 253 respectively a ratio of more than fifty to one. A model trained naively on this data learns to predict the common classes and quietly ignores the rare ones, which is the opposite of what matters clinically, since several of the rare classes are malignant. Reporting overall accuracy hides this failure entirely. The problem we address is therefore to build a model that performs well *across all eight classes at once* measured by balanced multi-class accuracy (BMA), the mean of the per-class recalls while respecting a tight parameter and compute budget and while making use of the clinical metadata that accompanies each image, including the large fraction of records where some of that metadata is missing.

1.4 Objective

The objectives of this work are:

1. To design a lightweight hybrid CNN Transformer classifier for ISIC-2019 that stays under 6M parameters and 1 GMAC and that fuses clinical metadata through a cross-attention head capable of handling missing fields without imputation.

2. To design a second, higher-capacity model in the same family a dual-backbone network with learned attention fusion and a Kolmogorov Arnold Network (KAN) classifier that trades footprint for a higher balanced accuracy.

3. To train both models, together with five established lightweight baselines, under one identical loss, augmentation, and evaluation pipeline so that the comparison is fair.

4. To quantify the contribution of each design choice (the CNN stem, the transformer trunk, the metadata head, and the dual-backbone fusion) through controlled ablations, and to report efficiency alongside accuracy.

1.5 Methodology in Brief

Every image passes through a fixed preprocessing pipeline: near-duplicate removal by perceptual hashing, a lesion-grouped stratified train/validation/test split, Shades-of-Gray colour constancy [1], and resizing to 224×224. The cleaned images are stored as a single memory-mapped array so that training does not pay a per-batch decoding cost. All models are trained with Class-Balanced Focal Loss [7], [14] to counter the imbalance, with AdamW, a cosine schedule with warmup, weight EMA, and eight-view test-time augmentation at inference. The lightweight model combines a truncated MobileNetV2 stem with a small ViT trunk and a metadata cross-attention head; the flagship model (DEKAN) fuses DenseNet-121 and EfficientNet-B0 features through learned attention before a transformer trunk and a KAN classifier. The primary metric throughout is BMA.

1.6 Scopes and Challenges

The scope of this thesis is the eight-class ISIC-2019 classification task and the two proposed architectures, evaluated against same-budget lightweight baselines on a held-out test split. The main challenges were the extreme class imbalance, the substantial missingness in the metadata (around 30% of age values, for example) and a hard hardware budget all experiments run on a single 16 GB GPU under Windows, which rules out multi-worker data loading and forces aggressive use of mixed precision and a memory-mapped dataset. We did not extend the study to external test sets or to multi-seed reporting; both are discussed openly as limitations in Chapter 5.

Chapter 2

Literature Review

2.1 Preliminaries

Before reviewing prior work it helps to fix a few ideas that recur throughout the thesis. A convolutional neural network builds up a representation of an image through stacked local filters, which makes it very good at capturing texture and short-range structure but, by construction, slow to relate distant parts of an image to one another. A transformer takes the opposite stance: it treats the image as a set of tokens and lets every token attend to every other through self-attention [9], [21], so global relationships are available from the first layer at the cost of weaker built-in spatial priors and a hunger for data. The *attention* operation at the heart of both transformers and our fusion modules computes a weighted average of "value" vectors, where the weights come from the similarity between a "query" and a set of "keys"; when the query comes from one source and the keys and values from another, the same mechanism becomes *cross-attention*, which is how we inject clinical metadata into the image representation. Finally, a Kolmogorov–Arnold Network (KAN) [25] replaces the fixed activation and linear weights of an ordinary layer with learnable spline functions on the edges of the network, a recent idea we evaluate as a classifier head.

2.2 Review of Existing Research

2.2.1 Convolutional backbones used in this work

The convolutional components of both our models are standard, well-understood backbones, chosen because their behaviour and pretrained weights are reliable. **MobileNetV2** [10] introduced the inverted residual block with a linear bottleneck, which keeps the network cheap by expanding to a higher-dimensional space only inside each block and using depthwise convolutions. Its early stages encode generic edges and textures very efficiently, which is why we use a truncated MobileNetV2 as the stem of our lightweight model. **EfficientNet-B0** [17] is the base model of a family obtained by compound scaling of depth, width, and resolution together; at roughly 5M parameters it is a strong accuracy-per-FLOP baseline and one of the two backbones in our flagship model. **DenseNet-121** [6] connects each layer to every subsequent layer, encouraging feature reuse and giving a rich, gradient-friendly representation; it is the second backbone in the flagship model and contributes a different inductive bias from EfficientNet. We also include **ResNet-18** [2], the residual-learning network that made very deep training practical, as a familiar reference point.

2.2.2 Transformers and CNN Transformer hybrids

The **Vision Transformer** [21] showed that a pure transformer, given enough data, can match convolutional models on image classification by splitting an image into patches and treating them as a token sequence. Its weakness on smaller datasets motivated hybrid designs that keep a convolutional front end. **MobileViT** [24] interleaves convolutions with transformer blocks to get a mobile-friendly model that still reasons globally, and **EfficientFormer** [23] shows that transformer-grade accuracy can be reached at MobileNet-like latency with a carefully chosen dimension-consistent design. We treat both as direct, same-spirit baselines, since a reviewer will reasonably ask how a new hybrid compares against existing lightweight hybrids rather than against pure CNNs alone. Our own design differs from these in where the convolution-to-transformer hand-off happens and, more importantly, in that the transformer's class token is used as a query that attends over clinical metadata.

2.2.3 Kolmogorov Arnold Networks

KANs [25] are a recent alternative to the multilayer perceptron, grounded in the Kolmogorov Arnold representation theorem, in which the learnable parameters are the coefficients of B-spline functions placed on the network's edges rather than scalar weights feeding fixed activations. The promise is more expressive, more interpretable units, sometimes at fewer parameters. KANs are still new and most evidence comes from small scientific-regression problems, so whether they help as a classifier head on a noisy, imbalanced vision task is an open question. We include a KAN head as the final classifier in our flagship model to explore whether its learnable splines add value at the output stage of a well-trained vision backbone.

2.2.4 Clinical metadata fusion

The work closest in spirit to ours is by Pacheco and Krohling [20], [22]. They showed first that adding patient clinical information measurably improves automated skin cancer detection [20], and then proposed an attention-based mechanism to combine images and metadata inside a deep model [22]. Gessert et al. [19], the basis of the top ISIC-2019 challenge entry, also fed metadata into an ensemble of EfficientNets. The key methodological point we adopt from this line of work is that missing metadata should be represented explicitly with a learned "missing" embedding rather than imputed with a fabricated value, so the model receives an honest "unknown" signal. Where we depart from prior work is in the backbone: Pacheco and Krohling fused metadata onto heavy CNNs, whereas we attach the same idea to a lightweight hybrid and to a dual-backbone hybrid, and we make an explicit efficiency claim about the result.

2.2.5 Skin lesion classification and ISIC benchmarks

Beyond the specific components above, automated dermoscopy has a large literature. Esteva et al. [5] demonstrated dermatologist-level melanoma classification with a single large CNN, and the ISIC challenges [4] have since standardised evaluation. The ISIC-2019 dataset is itself a union of several sources, most prominently HAM10000 [11] and BCN20000 [13], which is why source-aware splitting matters when building a fair evaluation. On the ISIC-2019 leaderboard [15], where the official ranking metric is balanced accuracy, the strongest published entries are large ensembles: the top entry (DAISYLab) reached a balanced accuracy of 0.636 with an ensemble of multi-resolution EfficientNets plus SENet-154, followed by entries around 0.61 and 0.59, again ensembles of EfficientNet and SE-ResNeXt variants. These numbers set a useful ceiling and a reminder that single, lightweight models operate in a different efficiency regime than the leaderboard's ensembles.

Most directly relevant to our experimental design, Aktan et al. [26] recently compared several leading deep learning architectures on this exact dataset, providing a clean, single-model reference for how standard backbones behave on ISIC-2019 under a common protocol. Our study extends that kind of controlled comparison in two directions: we add clinical metadata fusion to the comparison, and we report results under a tight efficiency budget, so that accuracy is always read together with parameter count and compute.

2.3 Summary of Key Findings

Three observations from the literature shape our design. First, hybrid CNN–Transformer models are a sensible match for dermoscopy because lesions are defined by both local texture and global shape, but the existing lightweight hybrids do not use clinical context. Second, metadata fusion clearly helps on this task, and the practical detail that decides whether it helps is how missing values are handled learned embeddings rather than imputation. Third, the best-published numbers on ISIC-2019 come from heavy ensembles, which leaves an open and useful niche for a single, small model that uses metadata to close part of the gap. This is precisely the gap the two models in this thesis are designed to fill.

Chapter 3

Requirements, Impacts and Constraints

3.1 Dataset and Data Requirements

All experiments use the ISIC-2019 training collection, a public set of 25,331 dermoscopic images labelled across the eight diagnostic classes listed in Chapter 1. The collection is itself a union of three sources HAM10000, BCN20000, and MSK which we infer from the image-ID prefix, because keeping track of the source matters for building an honest train/test split. Each image also carries clinical metadata: approximate age, sex, the general anatomical site of the lesion, and a lesion identifier that groups multiple images of the same physical lesion.

The defining property of this dataset, and the one that drives most of our design decisions, is its class imbalance. Table 3.1 shows the per-class counts for the full collection. The largest class, nevus, is more than fifty times larger than the smallest, dermatofibroma. Several of the rare classes (for example SCC and AK) are clinically important, so a model that simply learns the head of this distribution would be useless in practice. This is the central reason we report balanced multi-class accuracy rather than overall accuracy throughout.

Table 3.1: Class distribution of the full ISIC-2019 collection (before deduplication) and imbalance relative to the largest class (NV).

Code	Full name	Images	Ratio vs. NV
MEL	Melanoma	4,522	2.8×
NV	Melanocytic nevus	12,875	1×
BCC	Basal cell carcinoma	3,323	3.9×
AK	Actinic keratosis	867	14.8×
BKL	Benign keratosis	2,624	4.9×
DF	Dermatofibroma	239	53.9×
VASC	Vascular lesion	253	50.9×
SCC	Squamous cell carcinoma	628	20.5×

The metadata is genuinely incomplete: roughly 30% of age values, about 20% of sex values, and about 15% of anatomical-site values are missing, with the recorded ages centred near 54 years. Rather than fill these gaps with guessed values, our requirement is that the model treat “missing” as a first-class, learnable signal, which is reflected directly in the architecture (Chapter 4).

3.1.1 Functional Requirements

The functional requirements specify what the system must do, independent of how well it does it. They follow directly from the goal of a clinically usable, metadata-aware classifier and are summarised in Table 3.2.

The system must accept a single dermoscopic image and return a probability distribution over the eight ISIC-2019 diagnostic categories (MEL, NV, BCC, AK, BKL, DF, VASC, SCC), rather than a single hard label, so that the output is a ranked set of possibilities suitable for decision support. It must optionally accept the patient metadata that accompanies each image age, sex, and the general anatomical site of the lesion and fold that context into the prediction. Crucially, it must continue to function when any of those metadata fields is absent, representing “missing” as an explicit, learnable signal rather than imputing a fabricated value, because a large fraction of ISIC-2019 records are incomplete. The system must apply a fixed preprocessing pipeline (colour constancy and resizing to 224×224) to raw input before inference, and must report performance per class and as balanced multi-class accuracy (BMA) so that the behaviour on rare, often malignant, classes is always visible. Finally, the whole training and evaluation procedure must be reproducible from configuration files.

Table 3.2: Functional requirements of the proposed system.

ID	Requirement
FR1	Accept an RGB dermoscopic image and output a probability distribution over the eight ISIC-2019 classes.
FR2	Optionally accept patient metadata (age, sex, anatomical site) and use it as additional diagnostic context.
FR3	Handle any missing metadata field through a learned “missing” representation, without imputation.
FR4	Return a ranked list of candidate diagnoses with associated scores (decision support, not a final diagnosis).
FR5	Apply the full preprocessing pipeline (Shades-of-Gray colour constancy, resize/centre-crop to 224×224) to raw input.
FR6	Train, validate, and evaluate from declarative configuration files with a fixed random seed.
FR7	Report per-class recall, BMA, macro-F1, macro-AUC, overall accuracy, and a normalised confusion matrix.

3.1.2 Non-Functional Requirements

The non-functional requirements specify the quality attributes the system must satisfy the constraints on *how* it performs its function. They are the requirements against which the design in Chapter 4 is judged, and they are listed in Table 3.3.

The central non-functional requirement is efficiency: the lightweight model must stay under 6M parameters and 1 GMAC so that it can be deployed on the modest hardware found in a typical clinic or on a portable device. The delivered lightweight hybrid meets this comfortably at 3.98M parameters and 0.631 GMACs. Closely related is deployability the system must train and run within a 16 GB memory ceiling and portability, since the implementation must run on a standard PyTorch stack, including under Windows, where the data loader is restricted to a single worker process. On the quality-of-output side, balanced multi-class accuracy is the primary performance requirement, and the model must remain robust to the dataset's extreme imbalance without the rare-class collapse that a naively trained classifier would exhibit. Reproducibility (fixed seed, deterministic pipeline, regenerable figures) and maintainability (a modular codebase split into preprocessing, datasets, losses, models, training, and analysis) are required so that every reported number can be traced to its source. Two further requirements are ethical in nature: the reliance on BMA is itself a fairness requirement, forcing the rare malignant classes to count equally, and the prohibition on imputing missing metadata is an honesty requirement, ensuring the model is never given a fabricated clinical value.

Table 3.3: Non-functional requirements and how the design satisfies them.

ID	Requirement	How it is met
NFR1	Efficiency / footprint: lightweight model $\leq$ 6M parameters and $\leq$ 1 GMAC.	Lightweight hybrid: 3.98M params, 0.631 GMACs.
NFR2	Deployability: train and run within a 16 GB memory budget.	AMP, channels-last layout, gradient accumulation, memory-mapped dataset.
NFR3	Accuracy: balanced multi-class accuracy as the primary metric; beat same-budget baselines.	Hybrid 0.6081 BMA (best in sub-6M budget); DEKAN 0.6438.
NFR4	Robustness to class imbalance: no rare-class collapse.	Class-Balanced Focal Loss, weighted sampler; stable training curves (Ch. 5).
NFR5	Reproducibility: deterministic, seed-fixed, configuration-driven.	Single seed (42); figures regenerated from saved outputs (Appendix A).
NFR6	Maintainability: modular, importable codebase.	Separate preprocessing / datasets / losses / models / training / analysis modules.
NFR7	Portability: standard framework; runs under Windows with zero-worker loading.	PyTorch + `timm`; memmap dataset compensates for single-worker constraint.
NFR8	Fairness and honesty: rare malignant classes weighted equally; no fabricated metadata.	BMA metric; explicit learned "missing" embeddings.
NFR9	Latency: single-image inference fast enough for interactive use.	Low GMAC count (0.631) keeps per-image inference inexpensive.

3.2 Societal Impact

The intended impact of this work is a decision-support tool small enough to run where care is actually delivered, using the same contextual information a clinician already has. The implications of such a tool span several dimensions, ranging from structural access to care to the health, safety, legal, and cultural challenges inherent in clinical AI deployment.

At a structural level, this work addresses the widening gap between the volume of skin lesions requiring screening and the limited number of dermatologists available to evaluate them. This shortage is most acute in low-resource settings; in countries such as Bangladesh, specialist dermatological coverage is concentrated in major urban centres, leaving rural populations with little access to early screening. A model that is cheap to deploy and capable of running offline lowers the barrier to first-line screening exactly where specialist coverage is thin, thereby widening healthcare access rather than concentrating it further. Because the system is assistive, its realistic social role is triage flagging lesions that warrant referral rather than replacing the specialist judgement that remains scarce.

The clinical benefit of this triage role follows from the prognosis of skin cancer, which depends heavily on early detection; melanoma, in particular, is highly treatable when caught early. However, introducing such a tool into a healthcare workflow introduces critical safety considerations. The most dangerous failure mode of a skin-lesion classifier is a false negative on a malignant class. This risk is precisely why we optimise and report balanced multi-class accuracy rather than overall accuracy; a model that scored well on the common benign nevus class while missing rare malignant ones would be numerically impressive yet clinically unsafe. A secondary safety concern is automation bias the tendency of users to defer to a confident machine output. We mitigate this by designing the system to produce a ranked set of possibilities rather than a single diagnosis, and by positioning it explicitly as an aid to, not a replacement for, a clinician who remains in the loop for every decision.

From a regulatory and legal standpoint, any clinical deployment of this system would sit within a medical-device regulatory regime and require corresponding clearance before use on patients; the present work makes no claim to such clearance. Liability for a missed or incorrect diagnosis, the need for informed consent governing the use of patient data, and compliance with data-protection law are all legal questions that a real deployment must answer. By relying only on a public, de-identified research dataset and by framing the output as decision support, this study stays within the bounds of research use and leaves these clinical-deployment questions explicitly out of scope.

Finally, the success and equity of this tool are deeply tied to cultural and demographic context. A specific and important limitation of this study concerns skin-tone representation. ISIC-2019 is assembled largely from European populations, and dermoscopic collections in general over-represent fair skin. A model trained on such data cannot be assumed to generalise to the darker and more diverse skin tones of a South Asian population such as Bangladesh's. Deploying it there without revalidation on representative local data would risk systematically worse performance for the very patients it is meant to serve. We flag this as a genuine equity concern rather than a solved problem, and note it again among the limitations and future work as a requirement for external validation on a local, representative dataset. More broadly, the acceptance of AI-assisted diagnosis depends on cultural attitudes toward automation in healthcare and on patient trust, which reinforces the case for a transparent, assistive design rather than an opaque, autonomous one.

3.3 Environmental Impact

The environmental footprint of a deep learning project is best understood through two distinct phases: the one-off energy cost of training the models, and the recurring, compounding cost of running inference once deployed. While efficiency in machine learning is often framed purely as a deployment advantage, it is fundamentally a sustainability argument. Designing models that require fewer computational resources directly translates to reduced energy consumption and a lower carbon footprint across the software lifecycle.

To quantify the initial environmental cost, we looked at the footprint of our training phase. All experiments were conducted on a single workstation equipped with an NVIDIA RTX 4070 Ti Super GPU, which has a 285 W board capacity and an estimated average draw of roughly 0.25 kW under typical training loads. The carbon cost for a single training run can be calculated using the formula:

$$\mathrm{CO_2eq} = P_{\mathrm{avg}} \times t_{\mathrm{train}} \times \mathrm{EF},$$

where P_{avg} represents the average board power, t_{train} denotes the wall-clock training time, and EF is the grid emission factor. Because Bangladesh's electrical grid relies heavily on fossil fuels, it carries a relatively high official grid emission factor of approximately 0.62 kg CO_2/kWh.

Table 3.4 applies this framework to our two proposed models, utilizing the wall-clock times recorded in our training logs. When scaling this math across the entire project—which included five baselines, both proposed architectures, and various ablation variants totaling roughly a dozen full training runs alongside minor exploratory sessions—the total training energy falls on the order of tens of kWh. This amounts to a few tens of kilograms of CO_2eq, a total footprint roughly equivalent to a single short drive in a gasoline-powered car. Crucially, this is orders of magnitude lower than the massive carbon expenditure required to train the heavy model ensembles that typically dominate the top of the ISIC-2019 leaderboard.

Table 3.4: Estimated training energy and carbon for the two proposed models, assuming an average board power of 0.25 kW and a grid emission factor of 0.62 kg CO_2/kWh.

Model	Wall-clock (h)	Energy (kWh)	CO_2eq (kg)
Lightweight hybrid	**≈3**	0.75	0.47
DEKAN (flagship)	**≈9**	2.25	1.40

While training is a sunk cost, the recurring inference cost dominates a clinical tool's lifetime footprint once it is deployed in the field. This is where our architectural design choices offer the greatest sustainability dividends. Because the lightweight hybrid processes a single classification using only 0.631 GMACs, its energy consumption per inference is practically negligible. This compact profile allows the model to run locally on low-power edge devices, such as an NVIDIA Jetson Orin Nano, which handles vision-transformer workloads within a modest 7–25 W power envelope. This bypasses the need for a continuously active, server-class GPU that draws ten to twenty times more power.

Furthermore, keeping the model compact extends sustainability beyond mere electricity consumption; it allows the software to run effectively on affordable or preexisting hardware. This reduces the institutional demand for new high-end accelerators and helps mitigate the growing problem of electronic waste. It also removes the requirement for a constant, high-bandwidth data center connection. By contrast, the multi-resolution EfficientNet and SENet-154 ensembles that define the current state of the art demand a massive computational toll for every single image processed. By introducing a two-tier design, this study provides a clear efficiency-versus-accuracy choice, offering a low-footprint path that aligns clinical utility with the principles of green, resource-conscious machine learning.

An honest accounting requires acknowledging certain limitations in these measurements. We did not directly meter hardware power draw at the wall outlet; these figures are estimates derived from the GPU's rated power specifications and national-average grid data. Real-world values will inevitably fluctuate based on exact board utilization and local variations in the energy mix. These metrics are presented as a reliable order-of-magnitude estimate rather than an absolute measurement, which is entirely sufficient to substantiate the claim that these models operate at a fraction of the environmental cost required by standard leaderboard architectures.

3.4 Ethical Issues

The ethical considerations of this work directly informed its technical design, specifically regarding data privacy, algorithmic fairness, and transparency in clinical reporting.

Regarding data ethics and privacy, this study relies exclusively on the public ISIC-2019 dataset released by the International Skin Imaging Collaboration. No new human-subject data were collected and all images were fully de-identified at the source. Limited patient metadata specifically age, sex and anatomical site is utilized strictly as model inputs to improve diagnostic support. This data is handled in strict compliance with ISIC data-use terms and carries zero risk of patient re-identification.

Algorithmic fairness is addressed through two key technical choices. First, optimizing for balanced multi-class accuracy forces the model to treat rare, malignant classes with the same weight as common benign ones, preventing the majority class from skewing performance metrics. Second, missing metadata is handled via a learned "missing" embedding rather than statistical imputation, ensuring the model never trains on fabricated clinical values. However, a significant fairness limitation remains: the skin-tone representation bias inherent in the ISIC-2019 dataset, which is largely derived from fair-skinned European populations. This remains an unresolved equity issue that requires external validation on local, diverse populations before any real-world deployment.

Finally, to ensure responsible deployment, the system is explicitly designed as a collaborative decision-support tool rather than an autonomous diagnostic agent. By outputting a ranked list of differential possibilities, the architecture keeps a clinician firmly in the loop and mitigates automation bias. In the interest of scientific transparency—which is a safety obligation in medical AI we report our negative and inconclusive results with equal weight to our successes. Specifically, we openly disclose that the metadata benefit for the lightweight model remained unconfirmed on a single test-set evaluation and that the KAN classifier head yielded no performance gains over a standard linear head.

3.5 Project Management Plan

This project was executed by a team of five over three academic semesters, spanning from Spring 2025 to Summer 2026, under the supervision of Dr. Md. Ashraful Alam.

The work was structured into three sequential phases—Pre-Thesis 1, Pre-Thesis 2 and the Final Thesis with specific tasks mapped to the timeline shown in Table 3.5.

The initial phase focused on foundational research and data engineering, including a comprehensive literature review, problem formulation regarding data imbalance, and the construction of a robust preprocessing pipeline (encompassing perceptual-hash deduplication, lesion-grouped stratified splitting, color constancy, and a memory-mapped dataset). The second phase transitioned into model development, where five lightweight baselines were implemented under a unified training recipe, alongside the design of the lightweight hybrid model. The final phase involved developing and training the higher-capacity DEKAN architecture, executing controlled ablation studies, conducting rigorous performance analyses, and drafting the final thesis manuscript.

Table 3.5: Project schedule across the three thesis phases. A filled cell (■) indicates active development during that phase.

Stage	Pre-Thesis 1	Pre-Thesis 2	Final Thesis
Problem definition & literature review	■	□	□
Data acquisition & preprocessing pipeline	■	■	□
Baseline implementation (5 models)	□	■	□
Lightweight hybrid design & training	□	■	■
DEKAN design & training	□	□	■
Ablation studies	□	□	■
Evaluation, analysis & figures	□	■	■
Thesis writing & revision	■	■	■

To ensure accountability and fulfill individual contribution criteria, responsibilities were divided along the structural lines of the technical pipeline. Team synchronization was maintained through strict Git version control and weekly supervisor progress reviews.

From a resource and infrastructure perspective, all training was restricted to a single shared workstation (detailed in Section 3.1.4). Managing the entire project within a configuration-driven pipeline on one machine was a deliberate management choice; it completely eliminated environment drift, ensuring a mathematically fair comparison across all five models. Major project milestones successfully delivered include a fully deduplicated and stratified dataset, a verified baseline suite, a budget-compliant lightweight hybrid model, the flagship DEKAN model, a completed ablation matrix, and the final reproducible code appendix.

3.6 Risk Management

A small number of critical technical risks were identified at the project's outset as potential threats to validity. The experimental design explicitly incorporated mitigations for these challenges, transforming potential points of failure into deliberate engineering choices. The risk register in Table 3.6 records each identified risk, its pre-mitigation severity, the specific engineering solution applied, and its residual status.

The most insidious threat was data leakage. Because the ISIC-2019 dataset contains near-duplicate photographs of individual lesions, a naive random split would have allowed identical underlying lesions to appear in both the training and test sets, artificially inflating performance metrics. We eliminated this risk by implementing perceptual-hash (pHash) deduplication followed by a strict lesion-grouped, source-stratified split. Another major threat was rare-class collapse, where severe class imbalance could cause a model to maximize overall accuracy by simply ignoring rare malignant conditions. This was successfully countered by employing a Class-Balanced Focal Loss, utilizing a weighted sampler, and choosing Balanced Multi-class Accuracy (BMA) as our primary selection metric; training logs confirm that no such collapse occurred.

Table 3.6: Project risk register (L = Likelihood, I = Impact, graded as High/Medium/Low).

Risk	L	I	Mitigation	Status
Train/test leakage via duplicate lesion images	H	H	pHash deduplication + lesion-grouped, source-stratified split	Mitigated
Rare-class collapse under imbalance	H	H	Class-Balanced Focal Loss, weighted sampler, BMA selection metric	Mitigated
Overfitting of from-scratch transformer on ~17k images	M	M	Pretrained CNN stem, stochastic depth, RandAugment, Mixup, weight EMA	Mitigated
16 GB GPU memory ceiling	H	M	AMP, channels-last, gradient accumulation, memmap dataset	Mitigated
Windows data-loader deadlock	M	M	Zero-worker loading + memory-mapped array	Mitigated
Substantial metadata missingness	H	M	Learned "missing" embeddings, no imputation	Mitigated
Single-seed variance	H	M	Acknowledged; multi-seed reporting deferred	Open (future work)
No external-dataset validation	M	H	Acknowledged; external validation deferred	Open (future work)
KAN head failing to add value	M	L	Linear-head ablation retained and reported	Realised, handled

We also encountered strict hardware and environmental constraints, namely a 16 GB GPU memory ceiling and a known Windows OS data-loader deadlock. These infrastructure risks were circumvented by adopting Automatic Mixed Precision (AMP), a channels-last memory layout, gradient accumulation, and a memory-mapped, zero-worker dataset configuration that bypassed the OS multi-processing limitation entirely. Similarly, the risk of training on fabricated clinical values due to missing patient metadata was resolved by introducing learned "missing" embeddings rather than statistical imputation.

Two risks could not be fully resolved within the current scope and are carried forward as acknowledged limitations: all reported results are derived from a single experimental seed, and the architectures have not yet been evaluated on an independent external dataset. Consequently, true seed variance and out-of-distribution generalization remain open questions for future validation. Finally, the risk of the KAN classifier head failing to add value was realized during testing; however, this was handled transparently by maintaining and reporting the linear-head ablation variant rather than suppressing the negative result.

3.7 Economic Analysis

The economic viability of this work depends less on its modest, one-off development cost than on its recurring operational footprint. By designing a compact model, we achieve a decisive financial advantage over both cloud-hosted inference infrastructures and the heavy model ensembles that typically dominate research leaderboards.

On the development side, expenses are largely limited to a fixed capital investment in hardware—specifically, a single workstation (Section 3.1.4) costing approximately USD 1,800 to 2,500—alongside the research team's time over three academic semesters. The entire software stack is open-source and free. Furthermore, the direct electricity cost of model training is negligible; as calculated in Section 3.3, the entire study consumed on the order of tens of kWh. At a standard Bangladeshi commercial electricity tariff of roughly USD 0.08 per kWh, this translates to an expenditure of only a few US dollars.

The true economic differentiation occurs at the deployment stage. Because the lightweight hybrid model requires only 3.98M parameters and 0.631 GMACs, it can be served locally from inexpensive edge hardware. A device like the NVIDIA Jetson Orin Nano Super, which is purpose-built to run vision-transformer workloads, demands a one-off purchase price of approximately USD 249, draws only a few watts of power, and operates entirely offline. This offline capability provides a massive operational advantage in rural clinics where network connectivity is unreliable.

By contrast, relying on cloud-based GPU inference introduces substantial, recurring operational costs. Hosting even a modest T4-class instance incurs an on-demand charge of USD 0.15 to 0.81 per GPU-hour. Maintaining a continuously available endpoint costs hundreds of dollars per month before factoring in data egress fees or the complex legal and privacy implications of transmitting patient images off-site. Table 3.7 models a three-year Total Cost of Ownership (TCO) comparison under a plausible clinical usage scenario. The edge architecture is roughly an order of magnitude cheaper over its operational lifetime, a financial gap that widens drastically when compared to leaderboard ensembles that require enterprise-grade data center accelerators.

Table 3.7: Illustrative three-year Total Cost of Ownership (TCO) for model deployment. Cloud estimates assume a T4 instance at USD 0.50/GPU-hour active 8 hours per day; edge estimates reflect the hardware purchase price plus local electricity tariffs.

Option	Up-front (USD)	3-year running (USD)	3-year total (USD)
Edge (Jetson Orin Nano Super, ∼10 W)	249	∼9	∼258
Cloud (T4-class, 8 h/day on demand)	0	∼4,380	∼4,380

This stark cost asymmetry underpins the project's cost-benefit rationale. While the structural societal value of early lesion triage is difficult to monetize precisely, a USD 249 device amortized over thousands of screenings yields a negligible cost per patient. Identifying a single malignant lesion early generates massive economic and clinical savings by avoiding late-stage cancer therapies.

The narrow accuracy gap between our lightweight model (0.6081 BMA) and a heavy leaderboard ensemble ($\approx$0.636) is easily justified by the order-of-magnitude reduction in deployment barriers. For clinical environments where maximizing diagnostic accuracy warrants a larger operational budget, the flagship DEKAN model (0.6438 BMA) offers a higher-performance alternative within the same design paradigm. This architecture ultimately provides healthcare institutions with a flexible, principled choice between two distinct cost-performance tiers rather than forcing a one-size-fits-all deployment strategy.

Chapter 4

Proposed Methodology

This chapter describes the full pipeline: how images and metadata are prepared, the two proposed architectures, the loss that handles the class imbalance, and the training and evaluation protocol shared by every model in the study.

4.1 Preprocessing Pipeline

Raw dermoscopic images vary in size, illumination, and, because ISIC-2019 is a union of several archives, in the presence of near-duplicate photographs of the same lesion. The pipeline cleans all of this once, before any training, in three steps.

Near-duplicate removal. We compute a 64-bit perceptual hash (pHash) for every image and group images whose hashes are within a Hamming distance of 4 using union-find, keeping a single representative per group. This removed 1,263 near-duplicates, leaving 24,068 unique images. Removing duplicates before splitting is what prevents the same lesion from leaking between training and test.

Lesion-grouped stratified split. We split the data 70/10/20 into training, validation, and test sets. The split is stratified jointly on class and source (HAM/BCN/MSK) so each split has a representative mix, and it is grouped by lesion identifier so that every image of a given physical lesion lands in the same split. The result is 16,872 training, 2,400 validation, and 4,796 test images.

Colour constancy and memmap. Each kept image is corrected with Shades-of-Gray colour constancy [1] (Minkowski norm $p = 6$) to reduce the strong illumination differences between dermatoscopes, then short-side-resized and centre-cropped to 224×224. The entire cleaned dataset is written once to a single memory-mapped `uint8` array together with the labels and metadata. At training time this array is opened read-only and stays resident in the host's page cache, so after the first epoch a batch costs little more than a memory copy, the key optimisation that makes single-worker data loading on Windows fast enough.

4.2 Overview of the Proposed Models

We propose two models that share a common design: convolutional features, a transformer trunk for global reasoning, and a cross-attention head that folds in clinical metadata. They sit at two different points on the accuracy/efficiency curve. Figure 4.1 sketches both. The first, which we refer to as the *lightweight hybrid*, is built to stay under 6M parameters. The second, *DEKAN*, spends more capacity in exchange for higher balanced accuracy.

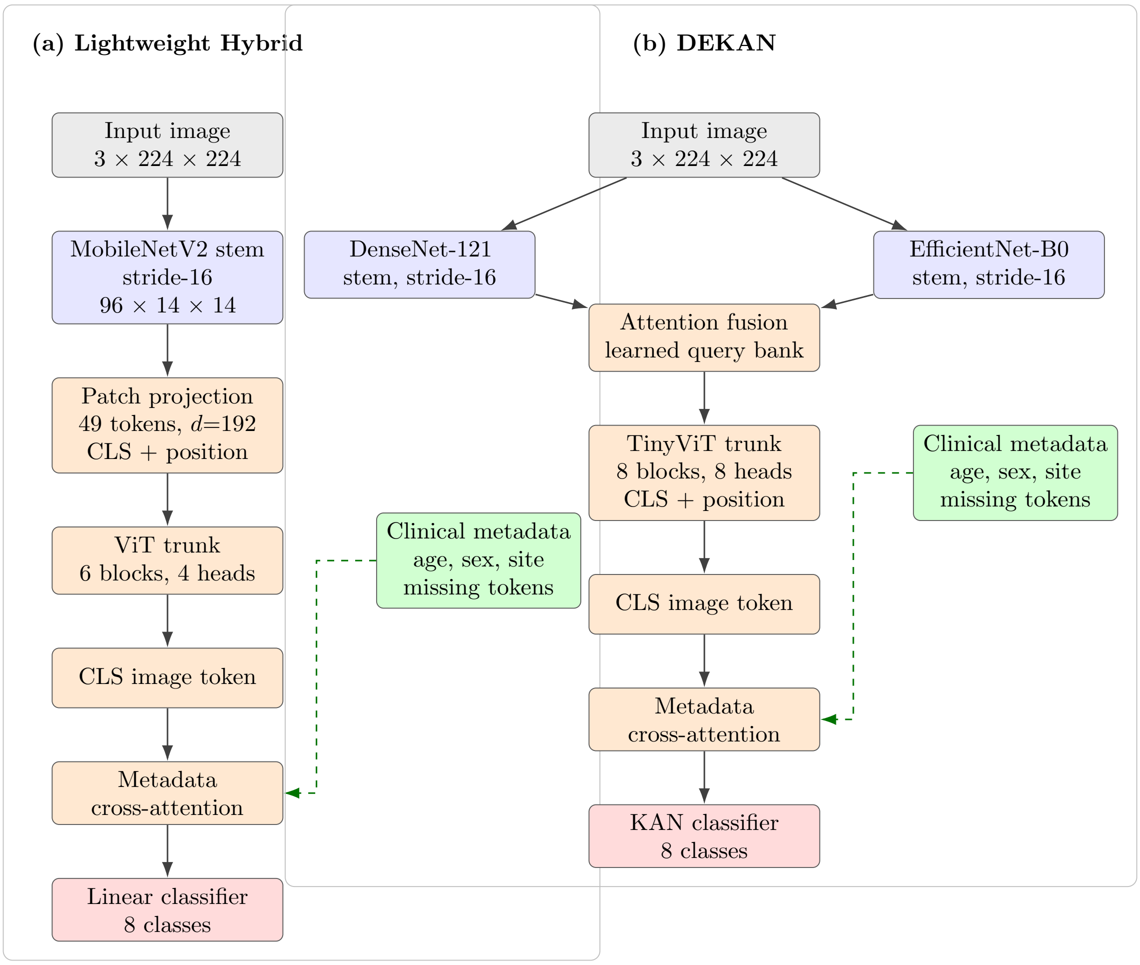

Figure 4.1: High-level architecture of the two proposed models. Both share the same philosophy convolutional features, a transformer trunk, and a metadata cross-attention head at two capacity points. (a) the lightweight hybrid uses a single truncated MobileNetV2 stem and a 6-layer ViT trunk with a linear head; (b) DEKAN fuses two pretrained backbones (DenseNet-121, EfficientNet-B0) through a learned attention bank, runs an 8-layer TinyViT trunk, and ends in a KAN head.

4.3 Lightweight Hybrid CNN–Transformer

4.3.1 CNN stem

The stem is a MobileNetV2 [10] (width 1.0), loaded with ImageNet weights and truncated at the stride-16 stage so that a 224×224 image produces a $96 \times 14 \times 14$ feature map. Using a pretrained convolutional front end gives the model generic edge and texture detectors for free, which matters because the transformer that follows is trained from scratch on only about 17,000 images. In our budget accounting the stem accounts for roughly 1.81M of the model's parameters.

4.3.2 Transformer trunk

A 2×2 stride-2 convolution projects the 96-channel, 14×14 feature map into a sequence of $7 \times 7 = 49$ tokens of dimension 192; this both embeds the patches and halves the spatial resolution so that self-attention, which is quadratic in the number of tokens, stays cheap. A learnable class (CLS) token is prepended and a learnable positional embedding added, giving 50 tokens. These pass through six pre-norm transformer blocks with four attention heads, an MLP expansion ratio of 2, and a linearly increasing stochastic-depth rate [3] up to 0.1, which is what keeps a small from-scratch transformer from overfitting at this data scale. The CLS token after the final layer summarises the image globally.

4.3.3 Metadata cross-attention head

Clinical metadata enters through a single cross-attention block in which the image CLS token is the query and three metadata tokens (age, sex, and anatomical site) are the keys and values. Age, a continuous value, is mapped to a token by a linear projection; sex and site, which are categorical, use learned embeddings. Missing values are handled explicitly: age has a dedicated learned "missing" vector that replaces its projection when the value is absent, and the categorical embeddings reserve an extra slot for "missing." Each token also carries a learned type embedding so the attention can tell the three fields apart. The block lets the model nudge the image representation using context, for instance, raising the prior on actinic keratosis for an older patient, without ever being forced to act on a fabricated value. The updated CLS token feeds a linear classifier producing eight logits.

The complete lightweight hybrid has 3.98M parameters and 0.631 GMACs, comfortably inside the budget of 6M parameters and 1 GMAC.

4.4 DEKAN: Dual-Backbone Flagship Model

DEKAN keeps the same overall shape but invests capacity where it buys accuracy.

Two complementary backbones. Instead of one stem it runs two in parallel

DenseNet-121 [6] and EfficientNet-B0 [17], both ImageNet-pretrained and both truncated to the stride-16 stage. The two networks encode the same lesion through different inductive biases, and the model is given the chance to use whichever is more informative per region.

Learned attention fusion. Each backbone's feature map is projected to 49 tokens of dimension 256 and tagged with a per-backbone embedding. A learnable bank of 49 query tokens then cross-attends over the concatenation of both token sets (98 tokens) and produces a single fused sequence of 49 tokens. This is a learned, content-dependent fusion rather than a fixed concatenation or average: the query bank decides, position by position, how much to take from each backbone.

Transformer trunk and metadata. A CLS token and positional embedding are prepended to the fused sequence, which then passes through an eight-layer TinyViT-style transformer trunk (dimension 256, eight heads, MLP ratio 4), followed by the same metadata cross-attention head described above.

KAN classifier. The final classifier is a single Kolmogorov–Arnold layer [25] mapping the 256-dimensional CLS representation to eight logits, with a grid size of 5 and cubic (order = 3) B-splines. Its spline basis is computed in full precision for numerical stability even under mixed precision. We deliberately keep a linear-head version of the otherwise-identical model so that the KAN's contribution can be measured directly (Chapter 5).

DEKAN has 16.45M parameters and 6.633 GMACs.

4.5 Class-Balanced Focal Loss

To counter the imbalance in Table 3.1, every model is trained with Class-Balanced Focal Loss, which combines the effective-number reweighting of Cui et al. [14] with the focal modulation of Lin et al. [7]. The effective number of samples for a class with n examples is $E(n) = (1-\beta^n)/(1-\beta)$, and the per-class weight is $\alpha = (1-\beta)/(1-\beta^n)$, normalised so the weights sum to the number of classes. The per-sample loss is

$$\mathcal{L} = -\,\alpha_y\,(1-p_y)^{\gamma}\,\mathrm{CE}_{\mathrm{smooth}}(\mathrm{logits}, y), \tag{4.1}$$

where p_y is the predicted probability of the true class, the $(1-p_y)^{\gamma}$ term down-weights easy examples so that learning focuses on the hard, often rare, cases, and $\mathrm{CE}_{\mathrm{smooth}}$ is a label-smoothed cross-entropy. In our configuration $\beta = 0.999$, $\gamma = 2.5$, and the label-smoothing factor is 0.1. The choice of $\beta = 0.999$ rather than a more aggressive value is deliberate: a larger β produced an extreme weight ratio between the common and rare classes and caused the rare classes to collapse early in training, whereas 0.999 gives stable optimisation.

4.6 Training Protocol

All models, including the five baselines and both proposed models, are trained with an identical recipe so that comparisons are fair; only batch size, gradient accumulation, and the backbone learning-rate scale differ, as dictated by each model's memory footprint. The optimiser is AdamW [16] with a base learning rate of 3×10^{-4}, $\beta = (0.9, 0.999)$, and gradients clipped to a maximum norm of 5. Pretrained CNN stems are fine-tuned gently at a fraction of the base learning rate (0.3× for the hybrid, 0.1× for DEKAN) while from-scratch components train at the full rate. The schedule is a five-epoch linear warmup followed by cosine annealing [8] to a minimum learning rate of 10^{-6}. Training uses automatic mixed precision and channels-last memory layout throughout, a weight exponential moving average (EMA, decay 0.9998) whose weights are used for validation and final evaluation, and a weighted random sampler that draws rare classes more often. Data augmentation combines horizontal and vertical flips, random resized crops, colour jitter, RandAugment [18], and Mixup [12] (strength $\alpha = 0.4$, applied to half the batches), with metrics always computed on the original, unmixed labels. Table 4.1 lists the per-model settings that vary.

Table 4.1: Per-model training settings. All other hyperparameters (loss, optimiser base LR, schedule, EMA, augmentation) are identical across models.

Model	Batch	Grad accum	Eff. batch	Epochs	Backbone LR
ResNet-18	256	1	256	100	0.3×
MobileNetV2	256	1	256	100	0.3×
EfficientNet-B0	128	2	256	100	0.3×
MobileViT-S	128	2	256	150	0.3×
EfficientFormer-L1	128	1	128	100	0.3×
Hybrid (full)	96	3	288	150	0.3×
DEKAN (full)	64	4	256	150	0.1×

4.7 Evaluation Protocol

The primary metric is balanced multi-class accuracy (BMA), the unweighted mean of the eight per-class recalls, which gives every class common or rare equal say. We additionally report macro-F1, macro-AUC, overall accuracy, and per-class recall, and we visualise errors with a confusion matrix normalised by true class. At inference we apply eight-view test-time augmentation (TTA): the model is run on the original image plus a fixed set of flips and rotations and the softmax outputs are averaged. The best checkpoint is selected by validation BMA and all headline numbers are reported on the held-out test split.

Chapter 5

Result Analysis

This chapter reports the results of the study. Unless stated otherwise, every number is measured on the held-out ISIC-2019 test split (4,796 images) at the eight-view test-time-augmentation (TTA) setting, using the checkpoint selected by the best validation BMA. All values are from a single run with seed 42.

5.1 Performance Evaluation

Table 5.1 compares the two proposed models against the five lightweight baselines, with every metric reported at the same TTA inference setting so the rows are directly comparable. The lightweight hybrid reaches a BMA of 0.6081 with only 3.98M parameters, which is the best balanced accuracy of any model in the sub-6M-parameter budget ahead of MobileViT-S (0.6020), its closest competitor, at fewer parameters and about two-thirds of the compute. The flagship DEKAN model reaches 0.6438 BMA among the main proposed models and also leads on every secondary metric (macro-F1 0.6059, macro-AUC 0.9271, accuracy 0.7467) at the cost of roughly four times the parameters and ten times the compute of the lightweight model. A linear-head ablation variant of DEKAN (dekan_linear) reaches 0.6510 BMA, which is the highest single-model figure in the study; the contribution of the KAN classifier relative to a plain linear head is analysed in Section 5.6.2. Figure 5.1 shows the same comparison graphically.

Table 5.1: Main results on the ISIC-2019 test set (with 8-view TTA). Best value in each column is in **bold**; the two proposed models are marked. A linear-head ablation of DEKAN (dekan_linear) reaches a higher BMA of 0.6510 but is reported with the other ablations in Section 5.6 rather than here, so this table is not its intended place; the bold BMA marks the best main model.

Model	Params (M)	GMACs	BMA	Macro-F1	Macro-AUC	Accuracy
ResNet-18	11.69	1.810	0.3368	0.2763	0.8217	0.5225
MobileNetV2	3.50	0.300	0.4738	0.4701	0.8827	0.6731
EfficientNet-B0	5.29	0.390	0.4946	0.4736	0.9037	0.6779
MobileViT-S	5.60	1.000	0.6020	0.5354	0.9218	0.6981
EfficientFormer-L1	12.27	1.300	0.5519	0.5779	0.9005	0.7333
Hybrid (ours, light)	3.98	0.631	0.6081	0.5602	0.9229	0.7162
DEKAN (ours, flagship)	16.45	6.633	**0.6438**	**0.6059**	**0.9271**	**0.7467**

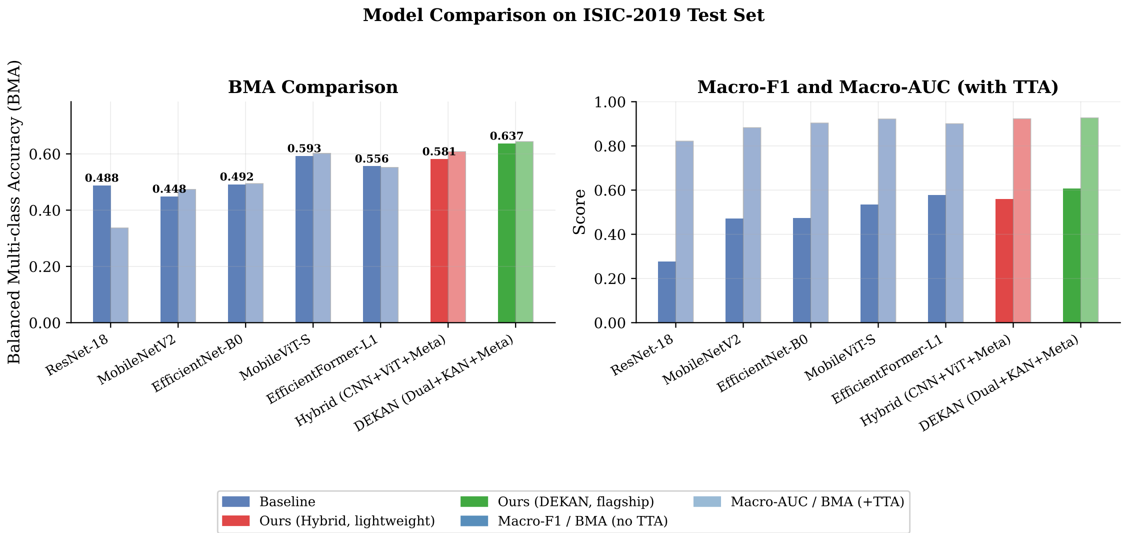

Figure 5.1: Model comparison on the ISIC-2019 test set: BMA (with and without TTA) on the left, macro-F1 and macro-AUC on the right. The two proposed models are highlighted.

The training behaviour of the lightweight hybrid is shown in Figure 5.2. Training and validation loss decrease smoothly and the validation BMA peaks at 0.6505 at epoch 149, with no sign of the rare-class collapse that a poorly tuned imbalance loss would cause evidence that the $\beta = 0.999$ choice in the Class-Balanced Focal Loss was appropriate.

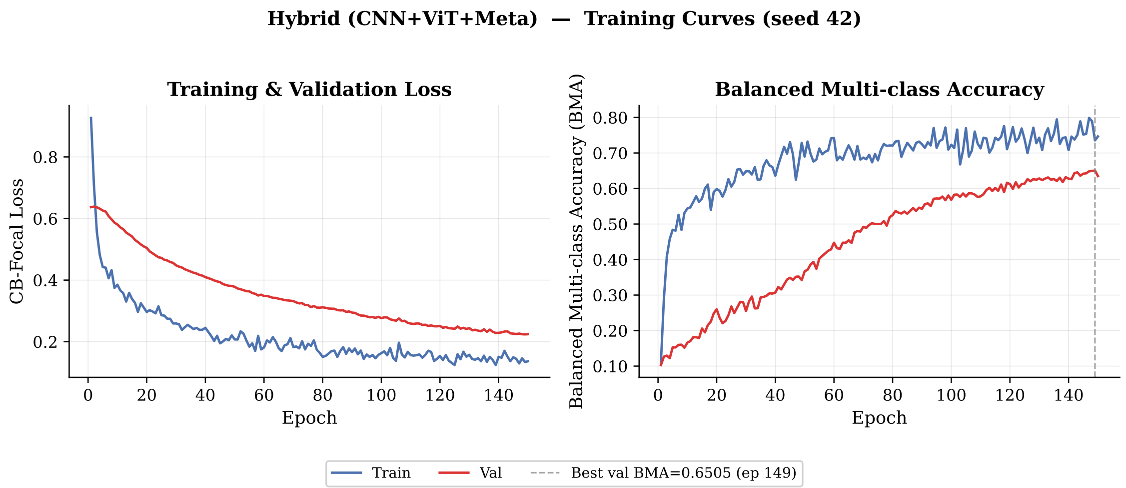

Figure 5.2: Training and validation loss (left) and BMA (right) for the lightweight hybrid model over 150 epochs. The dashed line marks the best validation BMA.

DEKAN follows a similar pattern, shown in Figure 5.3. Training and validation loss decrease smoothly over the full 150 epochs, and the validation BMA peaks late in training, again with no sign of rare-class collapse consistent with the loss configuration transferring well from the lightweight model to the higher-capacity dual-backbone setting.

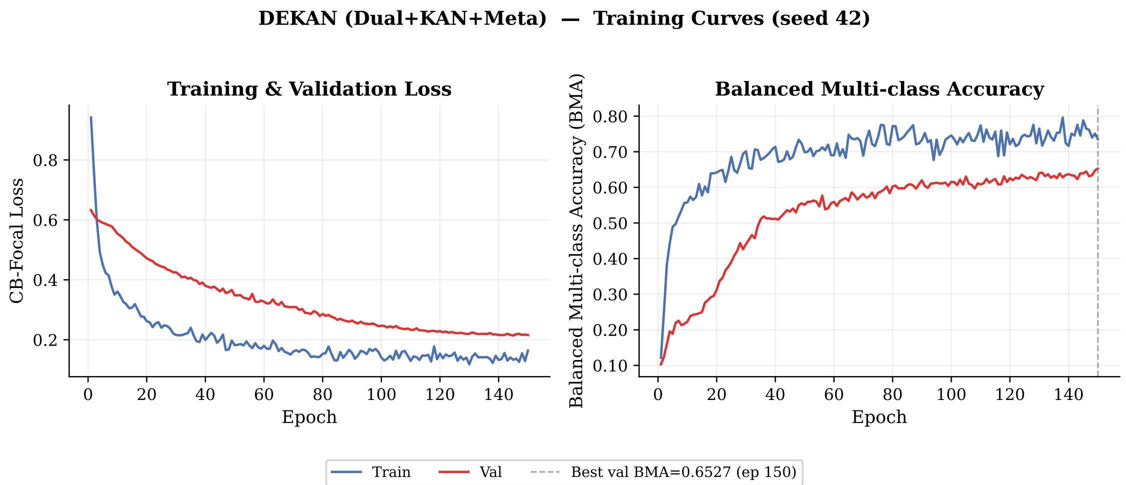

Figure 5.3: Training and validation loss (left) and BMA (right) for the DEKAN model over 150 epochs. The dashed line marks the best validation BMA.

5.2 Final Design Refinement

The final architectures presented in Chapter 4 are the direct result of iterative refinement, driven by extensive evaluation and ablation analysis. Several structural and procedural modifications were introduced throughout development to systematically address technical vulnerabilities identified during preliminary baseline experiments.

Early experimental trials revealed that severe class imbalance caused highly unstable learning behavior, which manifested as poor recognition of minority classes. To prevent rare-class collapse, the training framework was upgraded to incorporate Class-Balanced Focal Loss paired with a weighted sampler. This modification significantly stabilized optimization and improved balanced multi-class accuracy. Similarly, handling missing metadata posed an early challenge; conventional preprocessing methods required statistical imputation, which inadvertently introduced fabricated clinical values into the training loop. To preserve data integrity, imputation was replaced with a learned "missing" embedding strategy, enabling the network to explicitly represent absent information without generating artificial data.

Training stability also emerged as a significant barrier when optimization involved transformer-based components. To curb overfitting and enhance generalization on the constrained dataset, we integrated stochastic depth, weight Exponential Moving Average (EMA), Mixup augmentation, and RandAugment into the final execution pipeline.

System efficiency and OS-specific constraints required further engineering refinements. The Windows operating environment imposed strict synchronization limits on multi-worker data loading, creating a severe bottleneck that left the GPU underutilized. We bypassed this restriction entirely by building a custom memory-mapped dataset architecture coupled with a zero-worker loading strategy, ensuring high-throughput data delivery and strict reproducibility within our hardware constraints.

Finally, ablation experiments indicated that the much-hyped KAN classifier head did not consistently outperform a standard linear layer. Rather than filtering out this negative outcome, both variants were retained and documented to maintain empirical transparency.

Collectively, these refinements yield an architecture that balances diagnostic accuracy, computational efficiency, algorithmic fairness, and deployment readiness.

Table 5.2: Summary of empirical issues identified during development and the corresponding design modifications implemented in the final architecture.

Identified Issue	**Design Modification**
Rare-class collapse under imbalance	Class-Balanced Focal Loss and weighted sampling
Missing metadata values	Learned "missing" embeddings
Transformer instability and overfitting	Stochastic depth, Mixup, RandAugment, and weight EMA
Slow data loading / Windows deadlock	Memory-mapped dataset and zero-worker loading
Uncertain KAN head effectiveness	Linear-head ablation retained and reported transparently

5.3 Per-Class Analysis

Because BMA is an average over classes, it is worth looking at where the models succeed and fail per class. Table 5.3 lists per-class recall for two representative baselines and the two proposed models, and Figure 5.4 compares the lightweight hybrid against EfficientNet-B0 class by class.

Table 5.3: Per-class recall on the ISIC-2019 test set (with TTA).

Class	EfficientNet-B0	MobileViT-S	Hybrid (ours)	DEKAN (ours)
MEL	0.594	0.447	0.518	0.611
NV	0.857	0.890	0.859	0.859
BCC	0.546	0.467	0.618	0.731
AK	0.348	0.452	0.587	0.368
BKL	0.366	0.587	0.632	0.622
DF	0.245	0.660	0.566	0.509
VASC	0.843	0.882	0.863	0.922
SCC	0.157	0.430	0.223	0.529

The pattern is consistent with the headline numbers. The lightweight hybrid lifts several of the hard middle classes relative to the convolutional baselines BCC, AK, and BKL in particular which is where its global reasoning and metadata context help most. Notably, the hybrid's AK recall (0.587) is the highest of any model in the study, including DEKAN (0.368); actinic keratosis is a sun-damage-related precancerous condition disproportionately common in older patients at exposed anatomical sites, precisely the profile where the metadata cross-attention shifts the prior most reliably. Its weakest class is SCC (0.223), a rare class that is easily confused with AK and BKL. DEKAN spreads its recall more evenly and recovers SCC substantially (0.529) while also posting the strongest VASC (0.922) and BCC (0.731), which is the main reason its balanced accuracy is highest. Figure 5.5 shows the full confusion matrix for the lightweight hybrid; the off-diagonal mass sits mostly among the keratinocytic classes (AK/BKL/SCC), exactly the clinically difficult group.

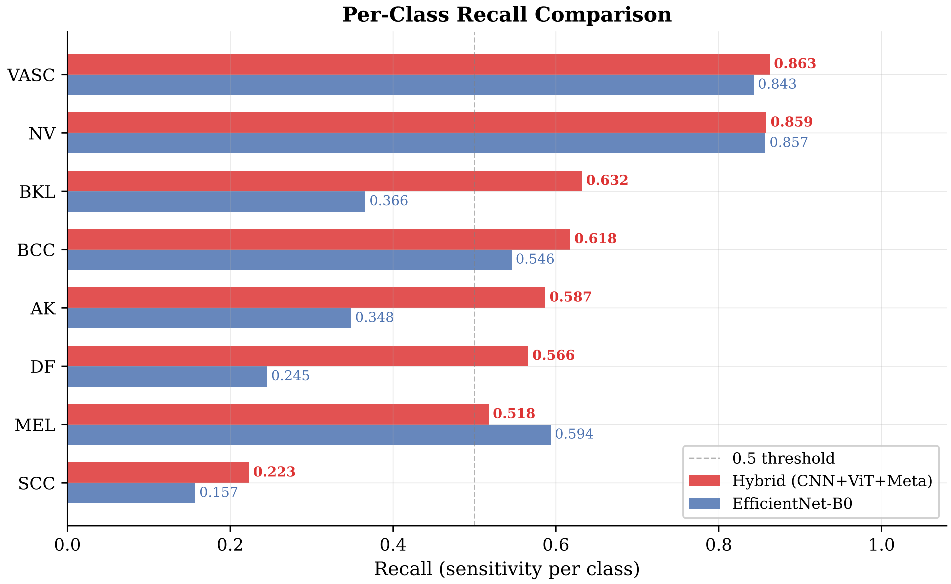

Figure 5.4: Per-class recall of the lightweight hybrid versus EfficientNet-B0, sorted by the hybrid's recall.

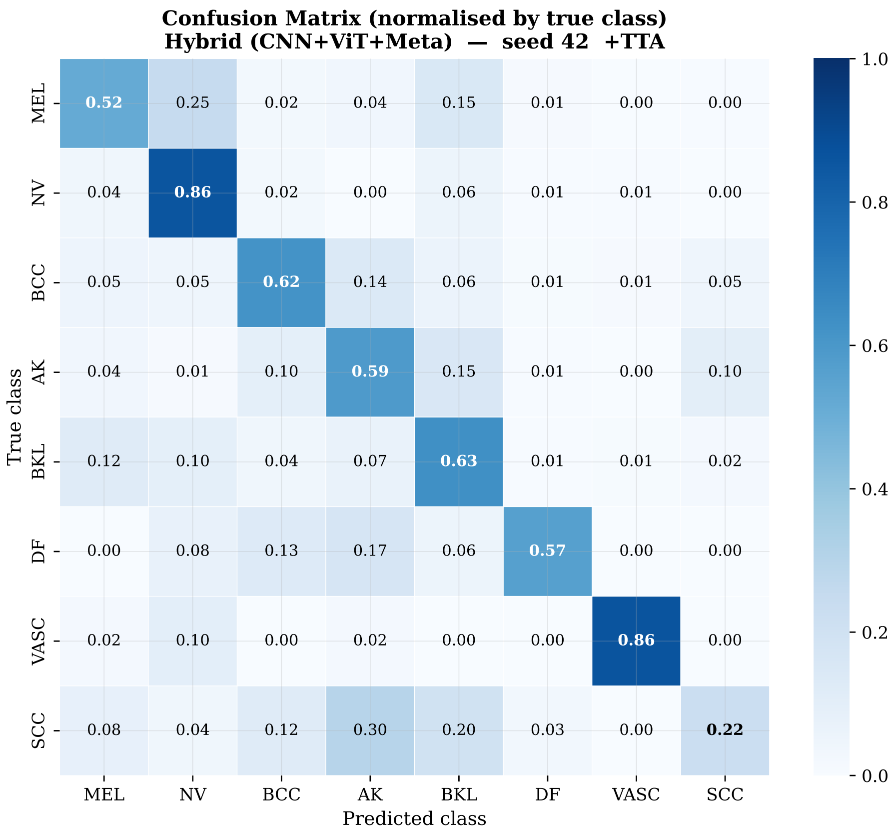

Figure 5.5: Confusion matrix for the lightweight hybrid (normalised by true class with TTA).

Figure 5.6 shows the same matrix for the DEKAN flagship. The keratinocytic confusion (AK/BKL/SCC) is reduced relative to the lightweight hybrid, and DEKAN recovers SCC substantially (0.529 vs. 0.223) the class that costs the hybrid most. VASC recall reaches 0.922, the highest of any model in the study.

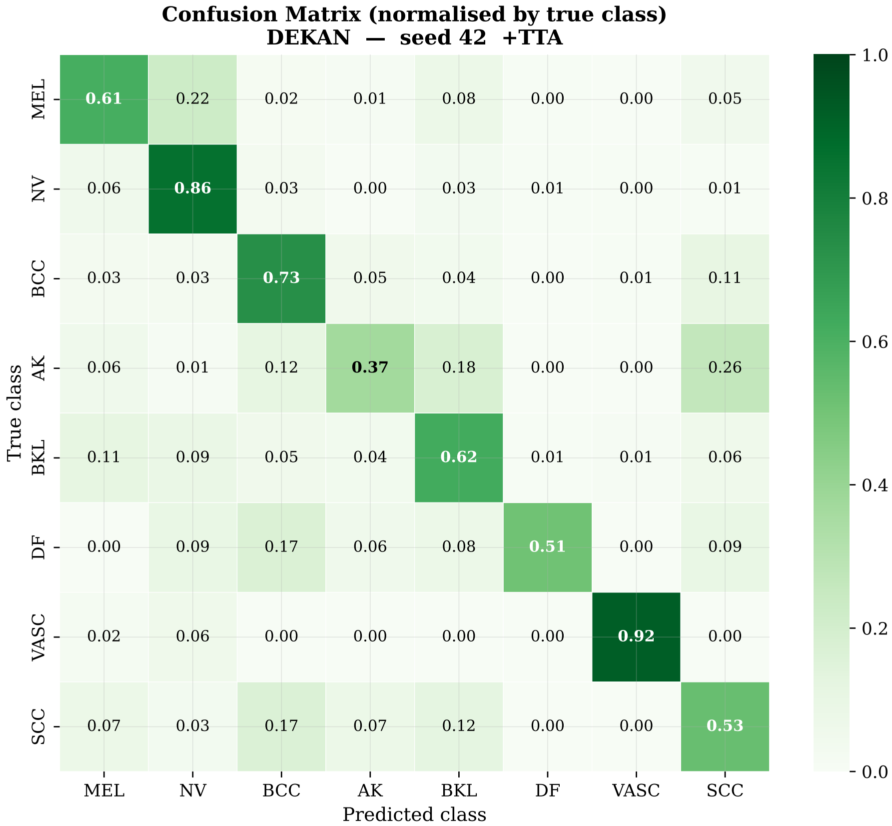

Figure 5.6: Confusion matrix for the DEKAN flagship model (normalised by true class, with TTA).

5.4 Efficiency Analysis

The point of the lightweight model is accuracy per unit of cost so Table 5.4 reports balanced accuracy per million parameters alongside the raw numbers and Figure 5.7 plots accuracy against parameter count with compute shown as bubble size. Among the full baselines and proposed models, the lightweight hybrid has the best BMA-per-parameter (0.153). The isolated CNN-stem (0.268) and pure-ViT (0.249) ablations score higher on this ratio alone, but both sit far below the full hybrid in absolute BMA, so their apparent efficiency is misleading parameter-efficiency without sufficient accuracy is not a useful operating point. The hybrid is therefore the best operating point that also clears a useful accuracy level. DEKAN sits at the opposite end: it buys the highest absolute accuracy with the lowest efficiency, which is the expected and intended trade-off between the two tiers.

Table 5.4: Efficiency: balanced accuracy per million parameters (TTA BMA).

Model	Params (M)	GMACs	BMA	BMA / M-param
ResNet-18	11.69	1.810	0.3368	0.029
MobileNetV2	3.50	0.300	0.4738	0.135
EfficientNet-B0	5.29	0.390	0.4946	0.093
MobileViT-S	5.60	1.000	0.6020	0.108
EfficientFormer-L1	12.27	1.300	0.5519	0.045
Hybrid (ours, light)	3.98	0.631	0.6081	**0.153**
DEKAN (ours, flagship)	16.45	6.633	0.6438	0.039

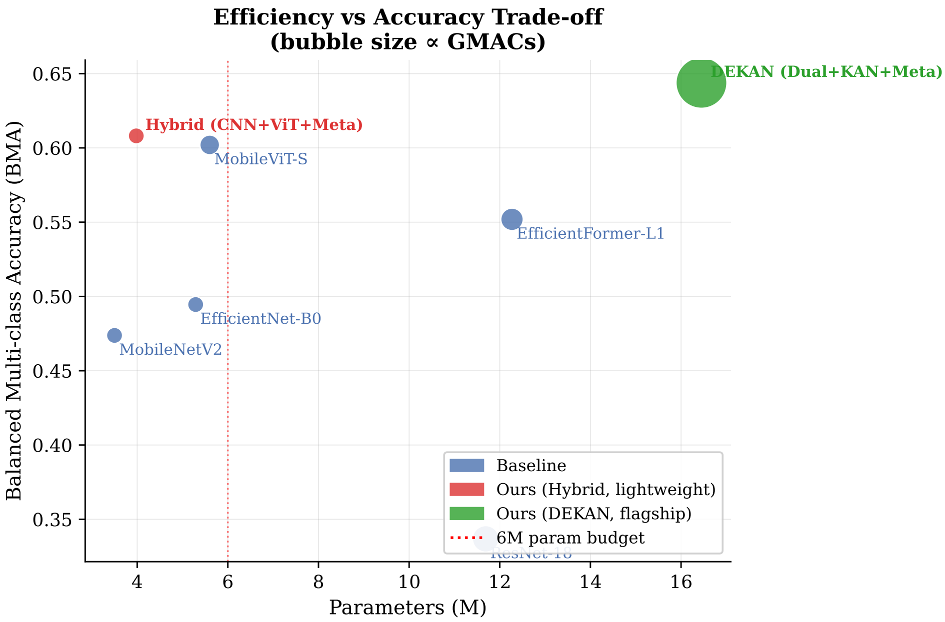

Figure 5.7: Accuracy vs. parameters; bubble size is proportional to GMACs. The dotted line marks the 6M-parameter budget. The lightweight hybrid (red) gives the best accuracy within budget.

5.5 Effect of Test-Time Augmentation

TTA helps most but not all models, so Table 5.5 reports BMA with and without it. The lightweight hybrid (full model) benefits most among the strong models (+0.027), and DEKAN gains a further +0.007. Among baselines, EfficientFormer-L1 drops slightly and ResNet-18 degrades sharply (−0.151); the ResNet-18 case is a known failure mode in which per-view predictions are confidently wrong in different directions, so averaging hurts.

The no-metadata ablation variant reveals a further pattern. Without the metadata head, the hybrid goes from 0.6216 to 0.6085 under TTA (−0.013) the same degradation direction as ResNet-18. Adding the metadata head reverses this: the full model goes from 0.5815 to 0.6081 (+0.027). The two variants converge to nearly the same TTA number (0.6085 vs 0.6081) from opposite sides. This TTA×metadata interaction suggests the cross-attention head stabilises per-view predictions; without it the model becomes overconfident in different directions across augmented views.

Table 5.5: BMA without and with 8-view TTA, and the change.

Model	BMA (no TTA)	BMA (TTA)	Δ
ResNet-18	0.4875	0.3368	−0.151
MobileNetV2	0.4485	0.4738	+0.025
EfficientNet-B0	0.4917	0.4946	+0.003
MobileViT-S	0.5925	0.6020	+0.010
EfficientFormer-L1	0.5559	0.5519	−0.004
Hybrid w/o metadata	0.6216	0.6085	−0.013
Hybrid (ours, light)	0.5815	0.6081	+0.027
DEKAN (ours, flagship)	0.6368	0.6438	+0.007

5.6 Analysis of Design Solutions (Ablations)

5.6.1 Lightweight hybrid

Table 5.6 and Figure 5.8 isolate the components of the lightweight model. The pure-ViT ablation (no CNN stem, 1.97M parameters) reaches 0.4908 BMA, virtually identical to the CNN-stem-only result (0.4853). The two components are roughly equal in isolation but strongly complementary when combined: adding the ViT trunk to the CNN stem yields a gain of +0.117 to +0.123 BMA, and the full hybrid reaches 0.6081. This is the primary architectural justification for the hybrid design neither local nor global features alone approach the performance the model achieves when both are present.

The effect of metadata in isolation is however modest at test time on this single run: the full model (0.6081) and the no-metadata variant (0.6085) are within 0.0004 of each other on the test set even though on the validation set the full model led by about 0.011 BMA. We therefore report the metadata contribution for the lightweight model honestly as inconclusive on a single test evaluation; the DEKAN results below give a clearer signal. The no-TTA numbers add a structural observation: without metadata the model scores 0.6216 before augmentation (versus 0.5815 for the full model), a gap of 0.040 that is entirely closed by TTA the full model gains +0.027 while the no-metadata variant loses −0.013 (Table 5.5). The two variants converge to the same TTA result from opposite sides, which points to the metadata head regularising per-view predictions rather than lifting the mean class probability.

Table 5.6: Lightweight hybrid ablation (TTA BMA). All four variants were trained and evaluated.

Configuration	Variant	BMA (TTA)
CNN stem + GAP + linear	hybrid_cnn_only	0.4853
Pure ViT, no CNN stem	hybrid_vit_only	0.4908
CNN + ViT, no metadata	hybrid_no_meta	0.6085
CNN + ViT + metadata (full)	hybrid_full	0.6081

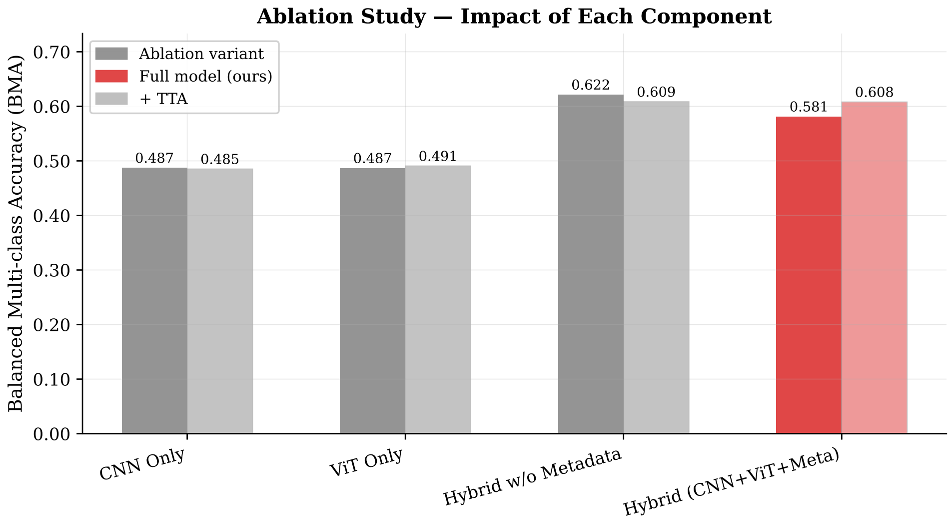

Figure 5.8: Lightweight hybrid ablation: BMA for all four variants. CNN-only and ViT-only perform similarly in isolation; the full hybrid is substantially higher.

5.6.2 DEKAN

Table 5.7 and Figure 5.9 isolate two effects in the flagship model: the contribution of clinical metadata and the choice of classifier head (KAN vs. linear).

Metadata. Comparing dekan_no_meta (0.6302) with dekan_linear (0.6510) both use a linear classifier head, differing only in the presence of the metadata cross-attention block shows a gain of +0.021 BMA. This is a clear and consistent signal that patient age, sex, and anatomical site provide useful diagnostic context at this model capacity.

KAN vs. linear head. Replacing the KAN classifier with a plain linear layer (dekan_linear, 0.6510) while keeping everything else identical produces a higher BMA than the full KAN model (dekan_full, 0.6438), a difference of +0.007 in favour of the linear head. The KAN classifier therefore does not improve over a plain linear layer on this task. This is an honest negative result: the KAN's learnable splines add no measurable value at the classifier stage of a well-trained vision backbone on this noisy imbalanced 8-class dataset. The likely explanation is that the discriminative difficulty on ISIC-2019 lies in the feature representation, not in the shape of the final decision boundary which a linear layer can already model adequately after the rich dual-backbone representation.

Table 5.7: DEKAN ablation: effect of clinical metadata and the KAN vs. linear classifier head (TTA BMA). The single-backbone variants were not trained, so the dual-backbone fusion gain cannot yet be quantified.

Configuration	Variant	BMA (TTA)
Dual backbone + fusion, no metadata, linear head	dekan_no_meta	0.6302
Dual backbone + fusion + metadata, linear head	dekan_linear	0.6510
Dual backbone + fusion + metadata + KAN (full)	dekan_full	0.6438

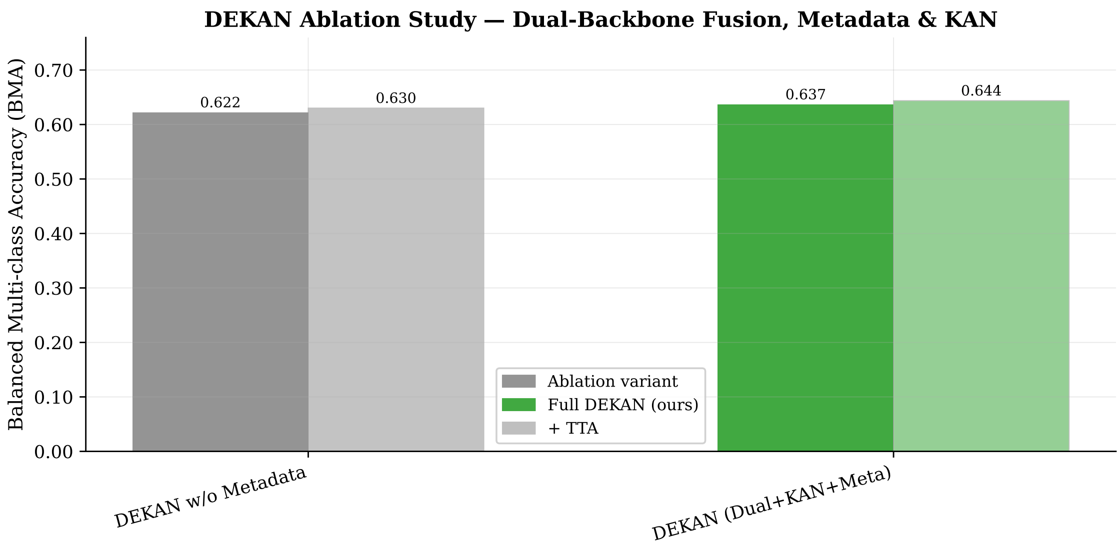

Figure 5.9: DEKAN ablation: effect of clinical metadata and KAN vs. linear classifier head on BMA. Single-backbone variants were not trained and are absent.

5.7 Comparisons and Relationships

Read against the wider literature, these results sit where one would expect a single-model, efficiency-constrained system to sit. The top entries on the ISIC-2019 challenge leaderboard [15] reach a balanced accuracy around 0.636 (DAISYLab) using large ensembles of multi-resolution EfficientNets and SENet-154. DEKAN, a single model, reaches 0.6438 BMA on our held-out split, and even the 3.98M-parameter lightweight hybrid reaches 0.6081 within striking distance of an entire ensemble at a tiny fraction of the cost. The comparison is not exact since the leaderboard is computed on the official challenge test server and our split is an internal one, so we present it as context rather than as a like-for-like ranking. Relative to controlled single-model comparisons on this dataset such as Aktan et al. [26], our contribution is to add clinical-metadata fusion to the comparison and to report accuracy together with a hard efficiency budget. The clearest relationship to prior work is with the metadata-fusion line of Pacheco and Krohling [20], [22]: we adopt their core idea of representing missing values with learned embeddings, and we extend their heavy-CNN setting to lightweight and dual-backbone hybrids.

5.8 Discussions

The results support the central claims of the thesis while leaving a few questions open. The lightweight hybrid does what it was designed to do: it beats all five same-budget baselines on balanced accuracy and is the most parameter-efficient model in its class which makes it a credible candidate for deployment on modest hardware. DEKAN shows that the same design scales: investing capacity in a second backbone, learned fusion and a deeper trunk pushes balanced accuracy to the level of a leaderboard ensemble in a single model. Metadata fusion helps clearly in the higher-capacity model and is at worst harmless in the lightweight one.

Two findings cut against initial expectations and are worth stating plainly. First, the benefit of metadata for the lightweight model could not be confirmed on a single test evaluation, even though the validation curve favoured it by 0.011 BMA. The DEKAN results give a clearer signal at higher capacity (+0.021 BMA between dekan_no_meta and dekan_linear), suggesting the metadata effect requires sufficient model capacity to manifest consistently. Second, the KAN classifier in DEKAN does not outperform a plain linear head: dekan_linear (0.6510) beats dekan_full (0.6438) by 0.007. This is the highest single-model BMA achieved in the study, but it belongs to the linear-head ablation, not to the KAN model, which is the intended flagship. The KAN's expressive spline activations appear to offer no advantage over a linear classifier once the dual-backbone representation is rich enough.

5.9 Limitations

Several limitations bound these conclusions. All numbers come from a single seed (42); publication-grade claims would require repeating each run over multiple seeds and reporting mean and standard deviation which we have not yet done. Two ablation variants were not trained dekan_densenet_only and dekan_effnet_only so the individual contribution of each backbone and the precise size of the dual-backbone fusion gain remain unquantified. We also did not evaluate on an external dataset so the generalisation of these models beyond the ISIC-2019 distribution is untested. Finally, the comparison with the ISIC-2019 leaderboard is contextual not a controlled head-to-head because the evaluation splits differ.

Chapter 6

Conclusion

6.1 Summary

This thesis set out to build skin-lesion classifiers for ISIC-2019 that are both accurate across all eight classes and practical to deploy, and that use the clinical metadata a dermatologist would normally have on hand. We approached the problem with a single design philosophy: convolutional features for local texture, a transformer trunk for global structure, and a cross-attention head that folds in age, sex, and lesion site, realised at two points on the efficiency curve. The lightweight hybrid, at 3.98M parameters and 0.631 GMACs, reaches a balanced multi-class accuracy of 0.6081 and is the most parameter-efficient model in the sub-6M budget, beating all five lightweight baselines including the strong hybrid baselines MobileViT-S and EfficientFormer-L1. The flagship DEKAN model, at 16.45M parameters, reaches 0.6438 BMA, comparable to the balanced accuracy of the top ISIC-2019 leaderboard ensemble in a single model; a linear-head ablation of the same architecture (dekan_linear) reaches 0.6510, the highest figure in the study, with the implication that the KAN classifier head adds no accuracy benefit on this task.

6.2 Contributions

The main contributions of this work are: (1) a lightweight hybrid CNN–Transformer with clinical-metadata cross-attention that handles missing fields with learned embeddings and meets an explicit budget of under 6M parameters and 1 GMAC; (2) DEKAN, a higher-capacity model in the same family that fuses two complementary pretrained backbones through learned attention and adds a Kolmogorov–Arnold classifier head; (3) a fair, single-protocol comparison of both models against five established baselines, with controlled ablations that quantify what each component adds, including the finding that neither a CNN stem nor a ViT trunk alone approaches the performance of their combination; and (4) two honest negative or inconclusive results: that the metadata benefit for the lightweight model could not be confirmed on a single test-set evaluation (though DEKAN gives a clear +0.021 BMA signal at higher capacity), and that the KAN classifier head in DEKAN does not improve over a plain linear head (dekan_linear at 0.6510 beats dekan_full at 0.6438 by +0.007).

6.3 Future Work

The most immediate next steps follow directly from the limitations in Chapter 5. Repeating every run across at least three seeds and reporting mean and standard deviation would put the comparisons on a publication-grade footing. Completing the two remaining unfinished ablations (DenseNet-121 alone and EfficientNet-B0 alone within the DEKAN framework) would pin down the individual contribution of each backbone and quantify the size of the dual-backbone fusion gain. Evaluating on an external dataset, for example mapping the eight-class output to a melanoma-versus-rest decision on a separate collection, would test generalisation beyond ISIC-2019. Finally, qualitative interpretability, using Grad-CAM on the CNN stem and attention rollout on the transformer trunk, would help confirm that the models attend to clinically meaningful regions, which matters for any tool intended to support, rather than replace, a clinician.

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